

BEAMS GENERATED BY GAUSSIAN QUASI-HOMOGENEOUS SOURCES[☆]

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An explicit expression is presented for the cross-spectral density function of the light in any cross-section perpendicular to the axis of a beam generated by a planar, steady-state, quasi-homogeneous source, whose intensity distribution and degree of spatial coherence are both gaussian. The result is used to discuss some properties of such a beam. It is found, in particular, that the ratio of the transvers (spatial) coherence length of the light to the beam width is the same for every cross-section perpendicular to the beam axis.

1. Introduction

We showed not long ago [1] that certain types of sources, known as quasi-homogeneous sources [2], can generate highly directional beams, although such sources are spatially rather incoherent in the global sense. This prediction has since been experimentally verified [3,4]. Radiation from quasi-homogeneous sources^{†1} has other interesting properties. For example, apart from a constant proportionality factor, the angular distribution of the radiant intensity is independent of the intensity distribution of the light across the source being entirely specified by its degree of spatial coherence [ref. [2], eq. (4.8)].

In view of these rather remarkable properties it is desirable to study more fully the nature of the beam

field generated by a quasi-homogeneous source. In this note we report results of an investigation for the case when the intensity distribution of the light across the quasi-homogeneous source and its degree of spatial coherence are both gaussian. Our analysis shows the manner in which the beam width and the transverse spatial coherence length of the light generated by such a globally incoherent source change with the distance of propagation. In particular, we find that the ratio of the spatial coherence length of the light in any plane perpendicular to the beam axis to the effective beam width in that plane is an invariant, i.e. this ratio is independent of the choice of the cross-section.

2. Propagation of the cross-spectral density function in a beam

Let σ be a planar source of any state of coherence, located in the plane $z = 0$ and radiating into the half-space $z > 0$. We assume that the fluctuations of the light generated by the source may be represented by a stationary statistical ensemble and we denote by

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^{†1} Since the publication of the paper cited in ref. [2] we found that the concept of a quasi-homogeneous source has a forerunner in an early investigation of Goodman [7]. In Appendix A of ref. [7], an approximate version of a basic theorem on radiation from quasi-homogeneous sources [eq. (4.5) of ref. [2] is derived [(Goodman's eq. 63)].

$W_\omega(x_1, y_1, z_1; x_2, y_2, z_2)$ the cross-spectral density function [5,6] at frequency ω of the light at two points (x_1, y_1, z_1) and (x_2, y_2, z_2) , with $z_1 \geq 0$, $z_2 \geq 0$. The cross-spectral density function may be expressed in the form of a "double angular spectrum of plane waves", viz., [ref. [5], eqs. (27) and (17)]:

$$W_\omega(x_1, y_1, z_1; x_2, y_2, z_2) = k^4 \iiint_{-\infty}^{\infty} \tilde{W}_\omega(k_{p_1}, k_{q_1}, 0; -k_{p_2} - k_{q_2}, 0) \times \exp[ik(p_1 x_1 + q_1 y_1 + m_1 z_1 - p_2 x_2 - q_2 y_2 - m_2 z_2)] \times dp_1 dq_1 dp_2 dq_2, \quad (2.1)$$

where

$$m_j = (1 - p_j^2 - q_j^2)^{1/2} \quad \text{if } p_j^2 + q_j^2 \leq 1 \\ = i(p_j^2 + q_j^2 - 1)^{1/2} \quad \text{if } p_j^2 + q_j^2 > 1, \quad (2.2)$$

($j = 1, 2$), and $\tilde{W}_\omega(f_{1x}, f_{1y}, 0; f_{2x}, f_{2y}, 0)$ is the four-dimensional spatial Fourier transform of the cross-spectral density function of the light in the source plane, defined as

$$\tilde{W}_\omega(f_{1x}, f_{1y}, 0; f_{2x}, f_{2y}, 0) = (2\pi)^{-4} \iiint_{-\infty}^{\infty} W_\omega(x'_1, y'_1, 0; x'_2, y'_2, 0) \times \exp[-i(f_{1x} x'_1 + f_{1y} y'_1 + f_{2x} x'_2 + f_{2y} y'_2)] \times dx'_1 dy'_1 dx'_2 dy'_2. \quad (2.3)$$

In (2.1) $k = \omega/c$ (c being the speed of light in vacuo) is the wave number associated with the frequency ω .

We will assume that the field generated by the source is a *beam*, propagated about the z -direction. In this case, only those plane waves propagating in directions that make small angles with the z -axis, i.e. for which $p_1^2 + q_1^2 \ll 1$, $p_2^2 + q_2^2 \ll 1$ contribute substantially to the integral in (2.3). This conclusion may be verified by a rigorous mathematical argument and may be shown^{†2} to be equivalent to the requirement that $|\tilde{W}(kp, kq, 0; -kp, -kq, 0)|$ is appreciable

only when $p^2 + q^2 \ll 1$. In this case one has from (2.2)

$$m_j \approx 1 - \frac{1}{2}(p_j^2 + q_j^2), \quad (j = 1, 2). \quad (2.4)$$

Accordingly we will make this approximation in eq. (2.1).

We will only consider the field in a typical cross-section $z = \text{constant} > 0$ and hence we put in (2.1) $z_1 = z_2 = z > 0$ and, to simplify the notation, we set

$$(x_1, y_1) \equiv \mathbf{p}_1, \quad (x_2, y_2) \equiv \mathbf{p}_2, \quad (2.5a)$$

$$W_\omega(x_1, y_1, z; x_2, y_2, z) \equiv W(\mathbf{p}_1, \mathbf{p}_2, z), \quad (2.5b)$$

$$(f_{1x}, f_{1y}) \equiv \mathbf{f}_1, \quad (f_{2x}, f_{2y}) \equiv \mathbf{f}_2, \quad (2.5c)$$

$$\tilde{W}_\omega(f_{1x}, f_{1y}, 0; f_{2x}, f_{2y}, 0) \equiv \tilde{W}^{(0)}(\mathbf{f}_1, \mathbf{f}_2). \quad (2.5d)$$

With an obvious change in the variables of integration, eq. (2.1) then reduces to

$$W(\mathbf{p}_1, \mathbf{p}_2, z) = \iint \tilde{W}^{(0)}(\mathbf{f}_1, \mathbf{f}_2) \exp[-(iz/2k)(f_1^2 - f_2^2)] \times \exp[i(\mathbf{f}_1 \cdot \mathbf{p}_1 + \mathbf{f}_2 \cdot \mathbf{p}_2)] d^2 f_1 d^2 f_2, \quad (2.6)$$

where, according to (2.3), in the notation indicated in (2.5),

$$\tilde{W}^{(0)}(\mathbf{f}_1, \mathbf{f}_2) = (2\pi)^{-4} \iint W(\mathbf{p}'_1, \mathbf{p}'_2, 0) \times \exp[-i(\mathbf{f}_1 \cdot \mathbf{p}'_1 + \mathbf{f}_2 \cdot \mathbf{p}'_2)] d^2 \rho'_1 d^2 \rho'_2. \quad (2.7)$$

In eq. (2.6) each of the two integrations is taken independently over the whole two-dimensional spatial frequency plane and in eq. (2.7) each is taken over the source plane $z = 0$.

For the purposes of later discussion we recall that in terms of the cross-spectral density function, the optical intensity (at frequency ω , not explicitly shown) at the point $(\mathbf{p}, z)$ is given by

$$I(\mathbf{p}, z) = W(\mathbf{p}, \mathbf{p}, z) \quad (2.8)$$

and that the complex degree of spatial coherence at that frequency, for light at the points $(\mathbf{p}_1, z)$ and $(\mathbf{p}_2, z)$, is defined by the formula [ref. [6], eq. (2.10)]

$$\mu(\mathbf{p}_1, \mathbf{p}_2, z) = \frac{W(\mathbf{p}_1, \mathbf{p}_2, z)}{[W(\mathbf{p}_1, \mathbf{p}_1, z)]^{1/2} [W(\mathbf{p}_2, \mathbf{p}_2, z)]^{1/2}}. \quad (2.9)$$

^{†2} This later condition follows, with the help of Schwarz' inequality, from the far-zone form of eq. (2.1) [given by eq. (34) of ref. [5]].

3. The cross-spectral density function of a light beam generated by a gaussian quasi-homogeneous source

Let us now assume that the source σ is a quasi-homogeneous source [2]. Such a source is characterized by the property that its degree of spatial coherence $\mu(\mathbf{p}_1, \mathbf{p}_2, 0)$ depends on $\mathbf{p}_1$ and $\mathbf{p}_2$ only through the difference $\mathbf{p}_1 - \mathbf{p}_2$ i.e. it is of the form

$$\mu(\mathbf{p}_1, \mathbf{p}_2, 0) = g^{(0)}(\mathbf{p}_1 - \mathbf{p}_2), \quad (3.1)$$

where $g^{(0)}(\mathbf{p}')$ is a function of a two-dimensional vector variable and that its optical intensity

$$I^{(0)}(\mathbf{p}) \equiv W(\mathbf{p}, \mathbf{p}, 0) \quad (3.2)$$

varies slowly with position ($\mathbf{p}$) across the source, remaining sensibly constant over linear dimensions of the order of the coherence length $l_g^{(0)}$ (the effective width of $|g^{(0)}(\mathbf{p}')|$). In addition we assume that the linear dimensions of the source are large compared with $l_g^{(0)}$.

The cross-spectral density function of the light across such a source may be approximated by [ref. [2], eq. (2.9)]

$$W(\mathbf{p}_1, \mathbf{p}_2, 0) = I^{(0)}[\frac{1}{2}(\mathbf{p}_1 + \mathbf{p}_2)] g^{(0)}(\mathbf{p}_1 - \mathbf{p}_2) \quad (3.3)$$

and its four-dimensional spatial Fourier transform is given by [ref. [2], eq. (3.9)]

$$\tilde{W}^{(0)}(\mathbf{f}_1, \mathbf{f}_2) = \tilde{I}^{(0)}(\mathbf{f}_1 + \mathbf{f}_2) \tilde{g}^{(0)}[\frac{1}{2}(\mathbf{f}_1 - \mathbf{f}_2)], \quad (3.4)$$

where $\tilde{I}^{(0)}(\mathbf{f})$ and $\tilde{g}^{(0)}(\mathbf{f}')$ are the two-dimensional spatial Fourier transforms of $I^{(0)}(\mathbf{p})$ and $g^{(0)}(\mathbf{p}')$ respectively, i.e.,

$$\tilde{I}^{(0)}(\mathbf{f}) = (2\pi)^{-2} \int I^{(0)}(\mathbf{p}) \exp(-i\mathbf{f} \cdot \mathbf{p}) d^2 p, \quad (3.5)$$

$$\tilde{g}^{(0)}(\mathbf{f}') = (2\pi)^{-2} \int g^{(0)}(\mathbf{p}') \exp(-i\mathbf{f}' \cdot \mathbf{p}') d^2 p'. \quad (3.6)$$

Suppose now that both $I^{(0)}(\mathbf{p})$ and $g^{(0)}(\mathbf{p}')$ are gaussian (in which case we will speak of a gaussian quasi-homogeneous source), i.e., are of the form

$$I^{(0)}(\mathbf{p}) = A \exp(-\rho^2/2\sigma_I^2) \quad (3.7)$$

$$g^{(0)}(\mathbf{p}') = \exp(-\rho'^2/2\sigma_g^2), \quad (3.8)$$

where A , σ_I and σ_g are positive constants. In order that the source be quasi-homogeneous we must have

$$\sigma_I \gg \sigma_g. \quad (3.9)$$

Upon taking the two-dimensional Fourier transforms of the expressions (3.7) and (3.8) and substituting the transforms into the formula (3.4) we find that

$$\begin{aligned} \tilde{W}^{(0)}(\mathbf{f}_1, \mathbf{f}_2) &= \frac{A}{(2\pi)^2} \sigma_I^2 \sigma_g^2 \\ &\times \exp[-\sigma_I^2(\mathbf{f}_1 + \mathbf{f}_2)^2/2] \exp[-\sigma_g^2(\mathbf{f}_1 - \mathbf{f}_2)^2/8]. \end{aligned} \quad (3.10)$$

Recalling the remarks that precede eq. (2.4) it readily follows from (3.10) that our gaussian quasi-homogeneous source will generate a beam provided that

$$\sigma_g \gg 1/k = c/\omega = \lambda/2\pi, \quad (3.11)$$

where λ denotes the wavelength of the light associated with the frequency ω .

Next we substitute from eq. (3.10) into eq. (2.6). The resulting multiple integral may be evaluated by a long but a straightforward calculation and leads to the following explicit expression for the cross-spectral density function of the light in any cross-section of the beam:

$$\begin{aligned} W(\mathbf{p}_1, \mathbf{p}_2, z) &= \frac{A}{[\Delta(z)]^2} \exp\left\{-\frac{(\mathbf{p}_1 + \mathbf{p}_2)^2}{8\sigma_I^2 [\Delta(z)]^2}\right\} \\ &\times \exp\left\{-\frac{(\mathbf{p}_1 - \mathbf{p}_2)^2}{2\sigma_g^2 [\Delta(z)]^2}\right\} \exp[i\varphi(\mathbf{p}_1, \mathbf{p}_2, z)], \end{aligned} \quad (3.12)$$

where

$$\Delta(z) = [1 + (z/k\sigma_I\sigma_g)^2]^{1/2} \quad (3.13)$$

and

$$\varphi(\mathbf{p}_1, \mathbf{p}_2, z) = \frac{z}{2k\sigma_I^2\sigma_g^2 [\Delta(z)]^2} (\rho_1^2 - \rho_2^2). \quad (3.14)$$

4. The nature of the beam generated by a gaussian quasi-homogeneous source

From eq. (3.12) we may readily deduce a number of results of physical interest. First we note that according to eqs. (2.8) and (3.12) the intensity of the light at any point ($\mathbf{p}, z$) in the half-space $z > 0$ is given by³

³ See next page.

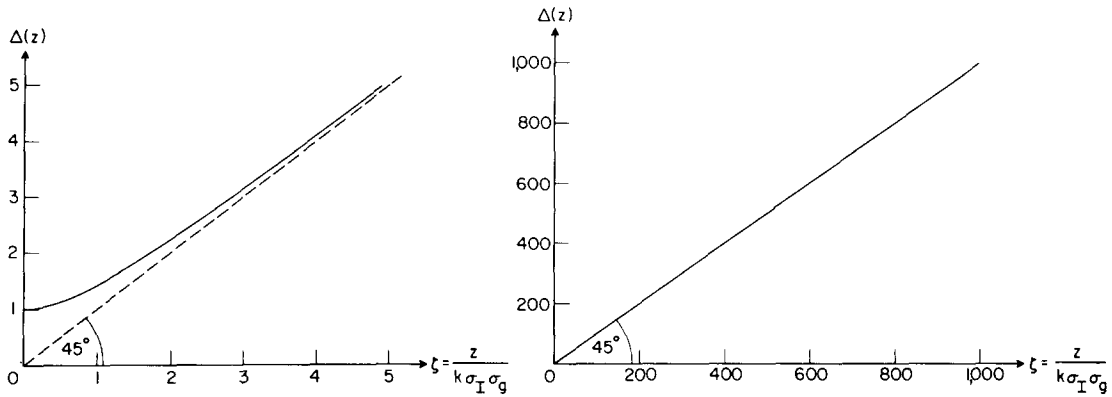


Fig. 1. The behavior of the expansion coefficient $\Delta(z) = [1 + (z/k\sigma_I\sigma_g)^2]^{1/2}$ as a function of the dimensionless variable $\zeta = z/k\sigma_I\sigma_g$.

$$I(\mathbf{p}, z) = \{A/[\Delta(z)]^2\} \exp \{-\rho^2/2\sigma_I^2[\Delta(z)]^2\}. \quad (4.1)$$

Equation (4.1) shows that in each plane $z = \text{const.} \geq 0$ the intensity is gaussian. On comparing (4.1) with (3.7) we see that on propagation from the source plane $z = 0$ to any plane $z = \text{const.} > 0$ the value of the intensity on the beam axis ($\mathbf{p} = 0$) decreases from the value A to $A/[\Delta(z)]^2$. Moreover, if we define the beam width $l_1(z)$ in any z -plane by the expression

$$l_1^2(z) = \int \rho^2 I(\mathbf{p}, z) d^2\rho / \int I(\mathbf{p}, z) d^2\rho, \quad (4.2)$$

where the integration extends over the particular z -plane, we readily find from eqs. (4.2) and (4.1) that

$$l_1(z) = \sqrt{2}\sigma_I\Delta(z). \quad (4.3)$$

According to eq. (3.13) $\Delta(0) = 1$ and we see that on propagation from the source plane to any particular z -plane the beam width has increased from the value $\sqrt{2}\sigma_I$ to $\sqrt{2}\sigma_I\Delta(z)$. For this reason we will refer to $\Delta(z)$ as the *expansion coefficient* of the beam. The behavior of $\Delta(z)$ as a function of the dimensionless variable $\zeta = z/k\sigma_I\sigma_g$ is shown in fig. 1.

The complex degree $\mu(\mathbf{p}_1, \mathbf{p}_2, z)$ of spatial coherence of the light in any cross-section of the beam can be determined by substituting for $W(\mathbf{p}_1, \mathbf{p}_2, z)$ from

(3.12) into (2.9). After a straightforward calculation we then find, if we also make use of the inequality (3.9), that

$$\mu(\mathbf{p}_1, \mathbf{p}_2, z) = g(\mathbf{p}_1 - \mathbf{p}_2, z) \exp[i\varphi(\mathbf{p}_1, \mathbf{p}_2, z)], \quad (4.4)$$

where $g(\mathbf{p}_1 - \mathbf{p}_2, z)$, the modulus of the complex degree of spatial coherence, is given by

$$g(\mathbf{p}_1 - \mathbf{p}_2, z) = \exp \{-(\mathbf{p}_1 - \mathbf{p}_2)^2/2\sigma_g^2[\Delta(z)]^2\} \quad (4.5)$$

and its phase $\varphi(\mathbf{p}_1, \mathbf{p}_2, z)$, is given by eq. (3.14). We see that the modulus of the complex degree of coherence of the light in any cross-section $z = \text{const.} \geq 0$ of the beam is gaussian. If we define the *transverse coherence length* $l_g(z)$ of the light in any cross-section by the expression

$$l_g^2(z) = \int \rho'^2 g(\mathbf{p}', z) d^2\rho' / \int g(\mathbf{p}', z) d^2\rho', \quad (4.6)$$

we readily find from (4.6) and (4.5) that

$$l_g(z) = \sqrt{2}\sigma_g\Delta(z). \quad (4.7)$$

Again, recalling that, according to (3.13) $\Delta(0) = 1$, we see that on propagation from the source plane to any particular z -plane the transverse coherence length of the light has increased from the value $\sqrt{2}\sigma_g$ to $\sqrt{2}\sigma_g\Delta(z)$.

It follows at once from eqs. (4.3) and (4.7) that

$$l_g(z)/l_1(z) = \sigma_g/\sigma_I, \quad (4.8)$$

i.e. the ratio $l_g(z)/l_1(z)$ is independent of z . We may express this result by saying that *in a beam generated by a gaussian quasi-homogeneous planar source the*

^{†3} The formulae (4.1) and (4.3) are in agreement with the corresponding expressions that follow in the quasi-homogeneous limit [characterized by eq. (3.9)] from recent results of Foley and Zubairy [8], concerning the intensity in fields generated by a class of sources of a more general type.

ratio of the transverse (spatial) coherence length of the light to the beam width is the same for every cross-section perpendicular to the beam axis.

Finally, for the sake of completeness, we also consider the behavior of the optical intensity and of the complex degree of spatial coherence in the far zone. For this purpose we note that when

$$z \gg k\sigma_g\sigma_1 \quad (4.9)$$

we have from (3.13)

$$\Delta(z) \sim z/k\sigma_1\sigma_g \quad (4.10)$$

and, with this approximation, (3.14) gives

$$\varphi(\mathbf{p}_1, \mathbf{p}_2, z) \sim k(\rho_1^2 - \rho_2^2)/2z. \quad (4.11)$$

On substituting from (4.10) and (4.11) into (4.1), (4.4) and (4.5) we obtain for the optical intensity and for the complex degree of spatial coherence of the light in the far zone the expressions^{†4} (denoted by the superscript ∞)

$$I^{(\infty)}(\mathbf{p}, z) = \frac{(k\sigma_1\sigma_g)^2 A}{z^2} \exp[-\frac{1}{2}(k\sigma_g)^2 \theta^2], \quad (4.12)$$

$$\begin{aligned} \mu^{(\infty)}(\mathbf{p}_1, \mathbf{p}_2, z) &= \exp[-\frac{1}{2}(k\sigma_1)^2/(\theta_1 - \theta_2)^2] \\ &\times \exp[ik(\rho_1^2 - \rho_2^2)/2z], \end{aligned} \quad (4.13)$$

^{†4} The inequality (4.9) is in agreement with the far zone criterion for the optical intensity derived by Leader [9]. However, in the present case this inequality is also seen to ensure that the far-zone formula (4.13) for the complex degree of spatial coherence applies.

where

$$\theta = \rho/z, \quad \theta_j = \rho_j/z, \quad (j = 1, 2). \quad (4.14)$$

The analysis presented in this note may readily be extended to beams generated by gaussian Schell-model sources^{†5}. We intend to discuss this generalization in another publication.

^{†5} For a definition of a Schell model source see, for example, ref. [8].

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